

TECHNICAL PROPOSAL: Reactor for Inertia Modulation via Resonant Field Exclusion (RIM-FE)

Author: [Your Name/Initials] Date: January 12, 2026 Core Thesis: Inertial mass is a consequence of the drag exerted by the Higgs vacuum (MET). This resistance can be locally eliminated by inducing a vacuum phase transition analogous to the Meissner effect in superconductors.

1. THEORETICAL FRAMEWORK AND INPUT DATA

1.1 Dynamic Mass (CERN Data)

Analysis of "Golden Events" (Mono-W topology, $R = 1.98$) demonstrated:

- Effective "Bubble" Mass: $M = 82.67 \pm 4.2$ GeV.
- Higgs Elasticity Coefficient (k_H): Mass increases by 1 GeV for every 7.5 GeV of environmental resistance (MET).
- Conclusion: Mass is not constant. If the interaction with the field is reduced ($MET \rightarrow 0$), the effective mass drops to its "bare" value (< 74 GeV, potentially towards zero for macroscopic objects).

1.2 Superconductivity Analogy (FermiLab Papers)

- Bafia et al. (2025): A "Frequency Dip" exists just below the critical point, signaling a change in vacuum coherence. Material purity is not ideal; "controlled disorder" is required for maximum Q-factor.
- Romanenko et al. (2014): Fast cooling (high dT/dt) across the critical temperature causes complete magnetic flux exclusion (Flux Expulsion) even in the presence of an ambient external field.

2. DEVICE DESIGN (HARDWARE)

The device is designed as a modified Superconducting Radio-Frequency (SRF) cavity.

2.1 Resonance Chamber (The Cavity)

- Material: Pure Niobium (Nb) doped with Nitrogen (N-doping) to achieve specific penetration depth and controlled disorder (as per Bafia et al., to capture the "Coherence Peak").
- Geometry: Elliptical 9-cell cavity, dimensionally scaled so that its EM Mode creates a standing wave with an effective field radius of $R \approx 1.98$ (in units relevant to the Higgs excitation wavelength).
- Surface Treatment: Electrochemical Polishing (EP) followed by "in-situ" baking at 120°C to optimize BCS resistance.

2.2 Field Generator

- **Frequency:** Must be tuned to a harmonic frequency corresponding to the 82 GeV energy state. (Note: Requires precise $E = h\nu$ conversion, likely in the THz range modulated on a GHz carrier wave).
- **Pulse Mode:** Capability to generate extreme rising edges (High Slew Rate) to simulate a "thermal shock."

3. OPERATIONAL PROTOCOL (MASS "SWITCH-OFF" ALGORITHM)

The process consists of three phases replicating flux exclusion in superconductors:

Phase 1: Tuning to "Higgs Dip" (Searching for the Dip)

- The device starts in a low-power mode.
- Frequency scanning is performed around the calculated resonance.
- **Goal:** To find the anomalous frequency drop (δf) identified by Bafia et al. as a precursor to the superconducting state. In our case, this indicates coupling with the Higgs field.

Phase 2: Pulse Expulsion (The Shock)

- This applies the principle from Romanenko et al. (Fast Cooling = Flux Expulsion).
- Instead of slowly increasing the field, a **high-energy RF pulse** with an extremely short rise time ($t < \tau_{relax}$, where τ is the relaxation time of the Higgs vacuum) is sent into the cavity.
- **Physical Phenomenon:** The rapid change in energy prevents the Higgs field from "soaking" into the field structure (Flux Trapping). The field is "startled" and expelled from the cavity volume.
- **Bubble Formation:** A region with $MET \approx 0$ forms around the cavity.

Phase 3: Maintaining "Coherence Peak"

- After the pulse, the system stabilizes at the "Frequency Dip" frequency.
- In this state, the cavity oscillates within the "coherence peak," where vacuum resistance is minimized.
- The body now exists in a state of reduced inertia.

4. EXPECTED RESULTS AND MEASUREMENT

4.1 Success Indicators

1. **Drop in Current Required for Oscillation:** If vacuum resistance (viscosity) drops, less energy will be needed to maintain cavity vibrations (Q-factor rises sharply above theoretical limits for standard superconductors).
2. **Gravimetric Anomaly:** If inertial mass is equivalent to gravitational mass (Equivalence Principle), a highly sensitive accelerometer inside the cavity should show anomalous reactions to external forces (acceleration will be higher than that corresponding to force $F = ma$).

4.2 Risks (Warning)

- **Flux Snapback:** If the field "snaps" (bubble collapse), the accumulated MET energy (50+ GeV per particle) is released instantaneously. This manifests as a strong burst of radiation (Bremsstrahlung) or mechanical impact.
- **Asymmetry:** According to Romanenko (2014), gradients must not form. If field expulsion is uneven, a thrust vector (propulsion) will be generated, which may be desirable but dangerously uncontrolled.

5. CONCLUSION

This proposal transforms passive matter into active material. By utilizing "Golden events" topology ($R = 1.98$, $M = 82$ GeV) and "fast cooling" technology for flux exclusion, we propose creating a shielded region of spacetime where inertia is a selectable parameter, not a constant.